



Research article

Analysis, control and design of a non-inverting buck-boost converter: A bump-less two-level T–S fuzzy PI control



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ABSTRACT

In this paper, a new modified fuzzy Two-Level Control Scheme (TLCS) is proposed to control a non-inverting buck-boost converter. Each level of fuzzy TLCS consists of a tuned fuzzy PI controller. In addition, a Takagi–Sugeno–Kang (TSK) fuzzy switch proposed to transfer the fuzzy PI controllers to each other in the control system. The major difficulty in designing fuzzy TLCS which degrades its performance is emerging unwanted drastic oscillations in the converter output voltage during replacing the controllers. Thereby, the fuzzy PI controllers in each level of TLCS structure are modified to eliminate these oscillations and improve the system performance. Some simulations and digital signal processor based experiments are conducted on a non-inverting buck-boost converter to support the effectiveness of the proposed TLCS in controlling the converter output voltage.

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1. Introduction

Nowadays, DC–DC converters are used with different input and output voltage-range requirements. For instance, a 3.3 V rechargeable Li-ion battery varies from 2.7 V to 4.2 V during discharge and charge. Also, solar systems output voltage varies in accordance with the atmospheric conditions as well as the solar radiation angle. To address these demands, an essential need is to develop converters with the ability of increasing and decreasing the input voltage to regulate the output voltage [1–4].

Several well-known topologies such as conventional buck-boost converter, Cuk converter, and Sepic converter have proposed to satisfy the aforesaid requirements. However, these converters suffer from high stress on switches or more specifically reversed output polarity in Cuk and conventional buck-boost converters.

A non-inverting buck-boost converter, as shown in Fig. 1, is a combination of a buck converter and a boost converter with two switches (S_1 and S_2), an inductor (L), and a capacitor (C). The extra switch and a diode are solved former converters problem. In addition, the non-inverting buck-boost converter is capable of being used in three separate operating modes which are buck mode (only S_1 is switched within one switching period and S_2 is opened),

boost mode (only S_2 is switched within one switching period and S_1 is kept close), and buck-boost mode (both S_1 and S_2 are switched concurrently) [5–7].

Generally, the linearized state-space averaging model is used to study the converters performance in different modes. In [8,9], this model have developed for buck and boost converters. Similarly in [5], the linear state-space averaging model for a non-inverting buck-boost converter have presented.

Based on these models, the open loop converter in boost mode has an unstable non-minimum phase characteristics, due to the right half plane zero. Thus, it is necessity to design a proper controller for stabilizing the converter. For this aim, usually a linear controller with constant coefficients designed based on the optimal converter performance in boost operation mode is used to control the converter in both buck and boost modes.

One of the main difficulties in controlling a non-inverting buck-boost converter is to provide a smooth transition from buck to boost operation and vice versa.

Many approaches have proposed a third mode of operation called transition mode to achieve a smooth transition between buck and boost modes [10–12]. In these cases, if the input voltage gets close enough to the output voltage, the converter will operate in transition mode. Otherwise, the converter operates in either buck or boost mode. In [5] and [13] the transition mode have achieved by complicated digital techniques ending to increase the circuit design complexity. In addition, the utilized buck-boost operation mode decreases the converter efficiency. Researcher in [14]

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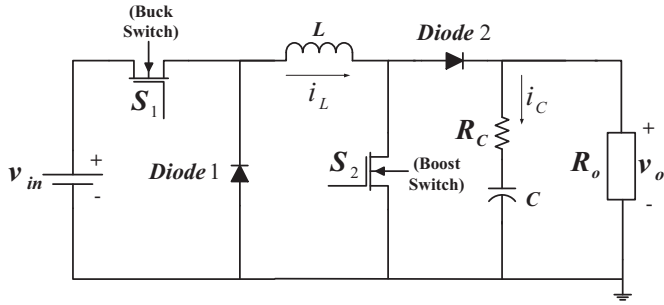


Fig. 1. Non-inverting buck-boost converter.

have proposed delta-sigma modulation technique to implement a transition mode; however it does not guarantee reaching to minimum ripple patterns. In [15], a current mode control with a constant time transition mode have suggested to improve the efficiency and alleviate the output voltage ripple. An extra control loop have introduced in [16], to dynamically change the operation modes between buck and boost. But the extra loop may possibly lead the system to instability. In [17], a double-carrier based sinusoidal pulse width modulation (SPWM) have proposed to achieve a smooth transition between the system modes, which is very difficult to be implemented in digital or analog control SPWM platforms.

Different controller methods have proposed to improve the transient response of a converter in average current mode control [18,19], current-programmed mode control [20–22], and voltage mode control [14]. But the influence of the input voltage as well as load disturbances on the output voltage cannot be fully eliminated. Following that, an input voltage feed-forward compensation approach have introduced in [23,24].

In aforementioned approaches a controller which is designed for the optimal performance of boost mode is used for controlling both buck and boost modes, which will deteriorate the converter performance in buck mode. Accordingly, using different dedicated controllers to each operation mode will increase the output voltage performance. In [25], a bump-less transfer algorithm proposed to transfer different velocities controllers to each other. In [26], predictive controllers have used with PI backup controllers to avoid any discontinuity in controlled variables. A switching control strategy for smooth transition of any two-degree-of-freedom controller while switching an active controller in closed-loop to another controller in a bank of inactive controllers have introduced in [27]. In this research work the nonlinearities are not considered, but recently switched fuzzy controller approaches have proposed by considering uncertainties and nonlinearities information in fuzzy membership functions to deal with parameter uncertainties in nonlinear systems [28,29].

In our previous work, we have introduced a classic multi-level controller to control a boost converter [30]. Consequently, since there are two distinct operation modes in a non-inverting buck-boost converter, in this paper, the classic multi-level controller idea is extended to a two-level fuzzy control scheme (TLCS) to control a non-inverting buck-boost converter. For each level of TLCS a fuzzy PI controller is designed using the obtained input-output data pairs from the corresponded PI classic controller for each level of TLCS. In this way, unlike the conventional control methods which are controlled the converter by only a single controller in all operating modes [5,21,31,32], two distinct fuzzy PI controllers are designed proportional to buck and boost modes. A fuzzy switch carries out the task of changing controllers to each other. The switch is designed based on expert knowledge using a TSK fuzzy system. Generally, while controllers in a system with different dynamics are switched to each other, drastic oscillations

are emerged in the output. This causes severe disruption in the non-inverting buck-boost converter operation. To prevent the oscillations, a modification is applied to the TLCS by adding a novel bump-less transition technique. As a consequence, the proposed method could efficiently improve the converter performance in both buck and boost modes while ensuring a smooth transition occurs between distinct operating modes.

The main contribution of this paper can be summarized as follows: (1) Each operation modes of a non-inverting buck-boost converter is controlled by a dedicated fuzzy PI controller to enhance the overall converter performance. (2) While a modified two-level control with a fuzzy switch changes the fuzzy PI controllers between each other in the control system no oscillation occurred on the converter output voltage. Finally, the efficiency of the proposed method is proved by both simulation and experimental results.

The rest of this paper is organized as follows. In Section 2, a brief review of the linearized state-space averaging model and its frequency analysis are presented, then following that pulse width modulation generator strategy is described. Section 3 describes the formulation of a TSK fuzzy system. The fuzzy TLCS is presented in Section 4. Some simulations and experiments are presented in support of the proposed TLCS performance in Section 5. Finally, conclusions are drawn in Section 6.

2. Linearized state-space averaged model and its analysis

2.1. Linearized state-space averaged model

In order to obtain the state-space model of a non-inverting buck-boost converter, both switching table and operating modes needs to be considered. Table 1 shows the converter switching patterns in different modes. Generally, the control system must be operated in such a way that buck functional state happens when the output voltage of the converter is lower than its input voltage. In this case, S_2 switch will be turned OFF during the switching interval and only S_1 switch will be switched ON and OFF. If the output voltage is greater than the input voltage, the converter must be changed into boost mode, in which S_1 switch remains constantly ON during the switching interval and S_2 is switched. If both S_1 and S_2 are switched simultaneously with the identical duty cycles, the non-inverting buck-boost converter will be similar to the conventional buck-boost converter in terms of performance; however, the polarity of the output voltage is positive and stresses on the switches as well as the diodes are remarkably reduced.

The important point is that since in the buck-boost mode both switches are continuously turned ON and OFF, the switching losses are higher than the other modes, which can reduce the converter efficiency. Thus, in this paper, the converter is controlled in a way that the buck-boost functional state is not being used at all. Avoiding from using this mode does not put any limitation on the converter functionality. In other words, depending on the amplitude of the input and output voltages, the converter can only be utilized in either buck or boost modes.

A small signal transfer function is achieved by obtaining the state-space averaging model and linearizing it around the

Table 1
Switching modes of a non-inverting buck-boost converter.

Mode	S_1	S_2
Buck	Switching	OFF
Boost	ON	Switching
Buck-Boost	Switching	Switching

operating point.

The transfer function of the control input ($\hat{d}$) and the source disturbance ($\hat{V}_{in}$) to the capacitor voltage in buck mode, Eq. (1), and boost mode, Eq. (2), are as follows [6]:

$$\hat{V}_c = \frac{\frac{V_{in}}{LC}}{s^2 + \frac{s}{R_o C} + \frac{1}{LC}} \hat{d} + \frac{\frac{D_1}{LC}}{s^2 + \frac{s}{R_o C} + \frac{1}{LC}} \hat{V}_{in} \quad (1)$$

$$\hat{V}_c = \frac{\frac{i_L}{C} \left(\frac{V_{in}}{L i_L} - s \right)}{s^2 + \frac{s}{R_o C} + \frac{(1-D_2)^2}{LC}} \hat{d} + \frac{\frac{(1-D_2)}{LC}}{s^2 + \frac{s}{R_o C} + \frac{(1-D_2)^2}{LC}} \hat{V}_{in} \quad (2)$$

where R_o , L and C are resistive, inductive and capacitive loads, respectively. D_1 and D_2 are the duty cycles of S_1 and S_2 switches; i_L is the inductor current, V_{in} represents the steady-state of input voltage and V_c shows the capacitor voltage. Moreover, the superscript $\wedge$ shows a small AC variation around the steady-state operating point and s is the Laplace variable.

Only the effect of the equivalent series resistance with capacitor (R_C) among the parasitic elements should be incorporated in transfer function. Because it creates a zero in the system transfer function. Therefore, the other parasitic elements because of their small values will not noticeably affect the transfer function parameters. Therefore, the other parasitic elements can be omitted from small-signal analysis of the circuit. However, they will be considered completely in both simulations and experimental tests.

Eq. (3) presents the relation between the output voltage (V_o) and the capacitor voltage.

$$\hat{V}_o = (1 + R_C C s) \hat{V}_c \quad (3)$$

By combining (3) with (1) and (2), and neglecting the input voltage dynamics, the transfer function of the control input (switch duty cycle) to the output voltage for buck and boost modes are expressed in (4) and (5) respectively [5].

$$\hat{V}_o = \frac{\frac{V_{in}}{LC} (1 + R_C C s)}{s^2 + \frac{s}{R_o C} + \frac{1}{LC}} \hat{d} \quad (4)$$

$$\hat{V}_o = \frac{\frac{i_L}{C} \left(\frac{V_{in}}{L i_L} - s \right) (1 + R_C C s)}{s^2 + \frac{s}{R_o C} + \frac{(1-D_2)^2}{LC}} \hat{d} \quad (5)$$

2.2. Frequency domain analysis

The Bode diagram is used to analyze different operating conditions of the converter. Accordingly to the open loop transfer functions (4) and (5), Fig. 2 shows the Bode diagrams for this converter in buck mode (a) and boost mode (b). As it can be seen, the stability margins are quite positive in buck mode, although the converter indicates negative phase and gain margins in the boost mode which make the converter completely unstable. The transfer function evaluation of the converter in boost mode shows an

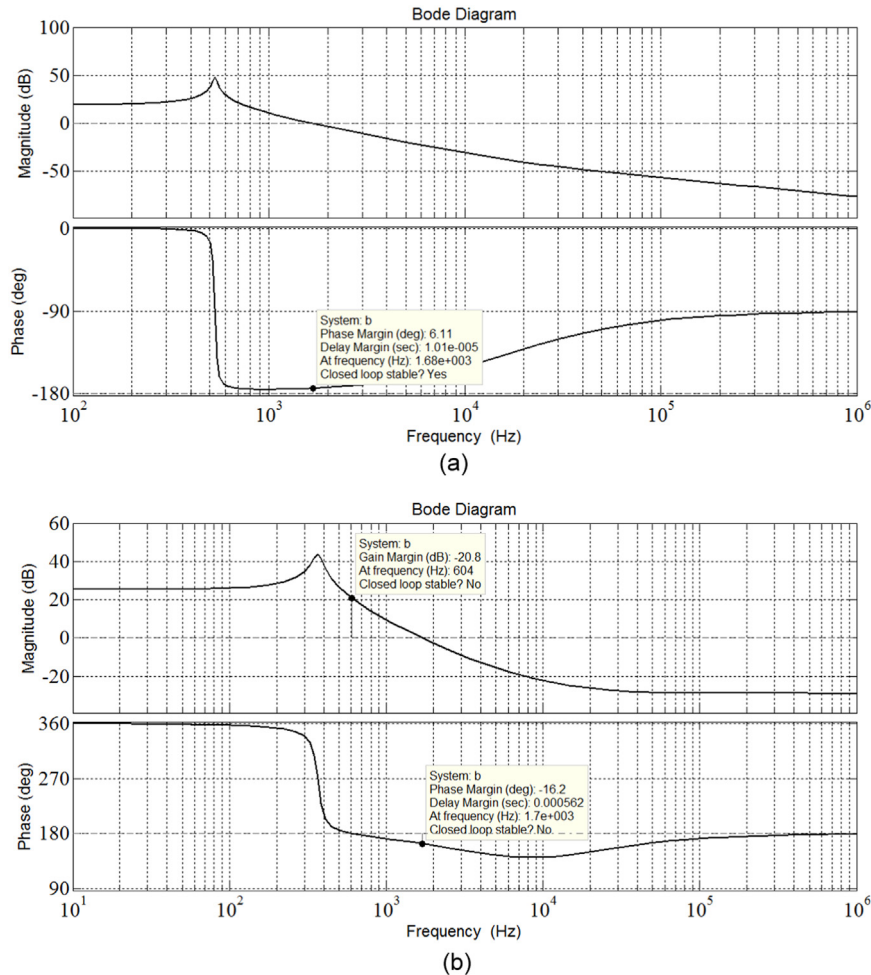


Fig. 2. Frequency response diagram of the non-inverting buck-boost converter, (a) Buck mode, (b) Boost mode.

unstable zero where the position depends on the inductor value, the inductor current, and the input voltage level. This unstable zero imposes some restrictions on the choice of a wide bandwidth in the controller design [33,34].

To sum up, the negative phase created by the right half plane zero prevents the designer to select a high cut-off frequency. This restriction slows dynamics of the system especially in buck operation mode.

2.3. Pulse width modulation

How to produce gate signals for controlling the converter switches with Pulse Width Modulation (PWM) is shown in Fig. 4 (b). It can be seen that PWM pulses are generated by intersection of a control signal (V_{ctrl}) and two carrier signals without any overlap to avoid the occurrence of converter buck-boost mode. The first triangular wave with the maximum amplitude (V_{H1}) and minimum amplitude (V_{L1}) and the second wave with the maximum amplitude (V_{H2}) and minimum amplitude (V_{L2}) determine the state of S_1 and S_2 , respectively. G_1 and G_2 are PWM signals driving S_1 and S_2 switches. As mentioned before, in this strategy by considering $V_{H1} = V_{L2}$ the buck-boost operation mode is discarded.

3. TSK fuzzy systems

Generally, fuzzy systems are universal approximators which means are able to approximate any unknown functions to any desired accuracy [35,36]. Consider a data set contains n data pairs. Each pair is made of a p -dimensional input-vector x and q -dimensional output-vector y . The number of fuzzy rules in Fuzzy Inference System (FIS) is r . Each typical fuzzy Mamdani rule i is shown as follows:

$$\begin{aligned} \text{IF } x_1 \text{ is } A_1^i \text{ and } x_2 \text{ is } A_2^i \dots \text{ and } x_p \text{ is } A_p^i \\ \text{THEN } y_1 \text{ is } C_1^i \dots \text{ and } y_p \text{ is } C_q^i \end{aligned} \quad (6)$$

where A_j^i and C_j^i are fuzzy sets that define an input and an output space partitioning.

In a TSK fuzzy system, the antecedent parts (IF) of IF-THEN rules are identical to the typical Mamdani fuzzy systems, but the consequent part (THEN) is a linear combination of the input variables [36]. Thus, a TSK fuzzy system does not require any defuzzification part. In fact, a TSK fuzzy system can be described as a mean weight of the obtained values from THEN parts of the rules. In a TSK fuzzy system, the consequent part of rule i for output j is computed as

$$y_j^i = b_{j0}^i + b_{j1}^i x_1 + b_{j2}^i x_2 + \dots + b_{jp}^i x_p \quad (7)$$

where b_j^i is the consequent coefficient of fuzzy rule i . Fuzzy Inference systems (FIS) is the most important part of a TSK fuzzy system which can be designed either based on an expert knowledge or input-output data pairs. Designing a TSK fuzzy system can be decomposed into two main phases: (1) generating a FIS and the associated antecedent part of fuzzy rules, (2) optimizing the coefficients of consequent part of fuzzy rules [36].

In expert knowledge based FIS design methods, an expert who has a strong knowledge about the system operation builds a FIS containing fuzzy rules. Unfortunately, selecting a large dispensable number of rules, not only leads to build a complicated fuzzy model and occupy large amount of memory, but also decreases the process speed. Conversely, choosing a small number of rules generates a weak fuzzy model which is not able to properly address the desired model. Moreover, the expert knowledge based FIS comes with lower accuracy in comparison with FIS designed based on the

input-output data pairs [37].

This is because the experts are only able to formulate their conscious human knowledge in form of a set of IF-THEN fuzzy rules and are unable to formulate their subconscious knowledge. Therefore, several methods have proposed to formulate both conscious and subconscious knowledge.

In these methods, the expert is considered as a black box and the input-output data pairs are collected. Indeed, both conscious and subconscious knowledge are hidden in a set of input-output pairs. Thus, by developing a fuzzy system based on the input-output pairs, in fact both knowledge types are being involved in FIS and fuzzy rules design.

Various fuzzy rule extraction methods from input-output data pairs have proposed. Some of the methods are as follows: self-organizing feature maps neural networks[38], hierarchical genetic algorithm [39], hunting search [40], and fuzzy clustering [41,42]. The main common drawback of these methods is that the clusters number, which is equivalent to the number of extracted rules number, needs to be pre-specified. For the aim of constructing a FIS to obtain a set of fuzzy rules that describes the system behavior, in this study a subtractive clustering algorithm is used, because it is not required any cluster numbers being determined in advance and the number of rules is obtained automatically [43]. The second phase of a TSK fuzzy system is determining the antecedent coefficient of the extracted fuzzy rules.

In a FIS generation using subtractive clustering each cluster center is a primary rule in the form of (6) that expresses the input-output of a system. In subtractive clustering, each data point is considered to be a potential cluster center. A measure of the potential is associated to each point according to its neighborhood and is defined by a radius, $r_a > 0$ known as the influence range of because it determines the neighborhood radius for each cluster center. For the point i , potential is written as

$$P_i = \sum_{j=1}^n e^{-4\|x_i - x_j\|^2 / r_a^2} \quad (8)$$

The point with the highest potential, P_1^* written x_1^* , is selected to become the first cluster center. Once the center is selected, the potential of each pair is decreased according to its distance to x_1^* . The new potential for pair i becomes

$$P_i = P_i - P_1^* e^{-4\|x_i - x_1^*\|^2 / r_b^2} \quad (9)$$

where $r_b > 0$ and equals to $r_b = r_a \times \eta$. The η is called a squash factor and represents measurable reduction in potential. The process is repeated until meet the algorithm termination criterion. To avoid introducing the noisy part of the data into the model, two thresholds named accept (ϵ^+) and reject (ϵ^-) ratios are considered. A new cluster candidate x_k^* at step k with its associated potential P_k^* , is managed as follows:

if $P_k^* > P_1^* \times \epsilon^+$, x_k^* is accepted as a new cluster center;
if $P_k^* < P_1^* \times \epsilon^-$, x_k^* is rejected as a new cluster center and the algorithm stops;
else
let d_{min} be the shortest distance between x_k^* and all previously found cluster centers;
if $(d_{min}/r_a) + (P_k^*/P_1^*) \geq 1 \times \epsilon^+$ is accepted as a new cluster center; (that means if it is far enough from the closest cluster)
else x_k^* is rejected, its potential is set to 0 and the algorithm goes on.

Unfortunately, there is no theoretical guidance for per-defined choosing subtractive clustering parameters. By the end of

clustering, a set of fuzzy rules will be obtained. Each cluster x_k^* represents a fuzzy rule. Given an input vector x , the degree of fulfillment of the antecedent clause of each rule i is defined as

$$\mu_{ij} = e^{-4 \|x_i - x_j\| / r_d^2} \quad (10)$$

where $\|x_i - x_j\|$ is the distance measure between then i -th data point and j -th data cluster center. Finally, the output of the TSK fuzzy system is computed as

$$\hat{y}_i = \frac{\sum_{j=1}^r \mu_{ij} \times y_j}{\sum_{j=1}^r \mu_{ij}} \quad (11)$$

The initial model is not optimal because the antecedent parts of rules were achieved only based on partitioning the input–output space. Thus, the model is not able to adequately portray the relation between input and output.

Several methods such as gradient descent and least mean squares by considering a Mean Square Error (MSE) criterion between the TSK fuzzy system output and the desired output regulate the coefficients of the consequent parts of the initial fuzzy rules. In this study, Recursive Least Square (RLS) method because of its higher speed and lower memory occupation comparing to other approaches is used to optimize the initial TSK fuzzy system [36,44]. A MSE criterion is defined as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n \|\hat{y}_i - y_i\|^2 \quad (12)$$

where $\hat{y}_i$ is the initial TSK fuzzy system output, y_i is the desired output and n is the total number of input–output pairs. Table 2 shows the parameters used in a subtractive clustering algorithm. For a thorough study of clustering methods, see references [43,45–47].

4. Non-inverting buck-boost converter control system

4.1. One-level PI controller

The non-inverting buck-boost converter is completely stable in buck mode. Thus, the desired response can be gained by selecting a wide bandwidth for a controller in this mode. However, due to the presence of the right half-plane zero and inappropriate stability margins in boost mode, the controller has to be designed with a narrow bandwidth that slows the converter dynamics in this mode. Therefore, using the boost mode characteristic to design a controller for controlling both operation modes will degrade the system performance in the buck operating intervals.

4.2. Two-level control scheme

The classic TLCS method solves the described problem in Section 4.1. By using two individual classic PI controllers for controlling different functional states of the non-inverting boost–buck converter. The two individual classic PI controllers for the buck mode (PI_{Buck}) and the boost mode (PI_{Boost}) of the converter are designed based on the well-known frequency response method [48]. In accordance with obvious difference between the converter

model in buck and boost modes, each classic PI controller is designed in such a way that the best time response performance is achieved for its operation mode.

The following equation shows a classic PI controller, which is frequently used in many industrial applications.

$$V_{ctrl}(t) = K_p e(t) + K_i \int_0^T e(\tau) d\tau \quad (13)$$

where $V_{ctrl}(t)$ is the controller output, $e(t)$ is the error signal generated from the difference between the reference signal and the system output voltage, and K_p and K_i are the proportional and integral gains, respectively. The schematic models of the classic and the fuzzy PI controllers are shown in Fig. 3.

Two fuzzy PI controllers are designed based on the method proposed in Section III by using the input–output pairs of the classic PI controllers.

The transition task between PI_{Buck} and PI_{Boost} controllers in both classic and fuzzy TLCSs is carried out by a fuzzy switch. At any time, the switch determines the converter functional mode in order to activate the proportional controller, i.e. PI_{Buck} or PI_{Boost} for the operating mode.

Finally, the controller output feeds into the PWM generator and the states of S_1 and S_2 switches are determined. The fuzzy switch description is presented in next section. A schematic model of the converter in presence of TLCS is shown in Fig. 4(a).

The parameters and time response characteristics of both classic and fuzzy PI_{Boost} and PI_{Buck} controllers are given in Table 3. In this table, t_r and t_s are rise-time and settling-time, respectively, and O.S. and U.S. respectively represent the over shoot and under shoot of the system step response in the presence of controllers. The table shows a significant difference between the classic and the fuzzy controllers. This difference will be significantly grown-up if higher-order controllers utilize.

At the time of transferring from the PI_{Buck} controller to the PI_{Boost} controller, where the ratio of output voltage to input voltage becomes greater than one, drastic oscillations appear in the output voltage. These oscillations hamper the positive features of the proposed method to show up. These oscillations are also emerged while a PI_{Boost} controller is changed with a PI_{Buck} controller, where the ratio of the output voltage to the input voltage becomes less than one.

These fluctuations are the consequence of the two controllers successive switching at instants of mode transition called changing points. In these points the output voltage is equal to the input voltage. The comparison of the one-level PI controller and the classic TLCS is illustrated in Fig. 5.

In Fig. 5, V_{in} and V_{ref} are input and reference voltages, respectively. Also, V_{O1} and V_{O2} represent the converter output voltages in the presence of the one-level controller and the classic TLCS, respectively.

Since the buck controller contains faster dynamic than the boost controller, large unintended oscillations are emerged at the changing points of the controllers. This problem not only weakens the performance of TLCS in comparison with the one-level controller, but also increases the switching losses and generating considerable Electro-Magnetic Interference (EMI). Fig. 5 proves the necessity of modification in a TLCS structure to eliminate the output voltage oscillations for achieving its desired performance.

4.3. Fuzzy switch

A fuzzy switch is defined to determine whether the PI_{Buck} or the PI_{Boost} must be taken into account. The fuzzy switch must activate the classic or the fuzzy PI_{Buck} controller at the changing point when the output voltage is less than the input voltage. Conversely,

Table 2
Parameters of sensitivity analysis.

Parameter	Value
Range of influence	1
Squash factor	2
Accept ratio	0.5
Reject ratio	0.15

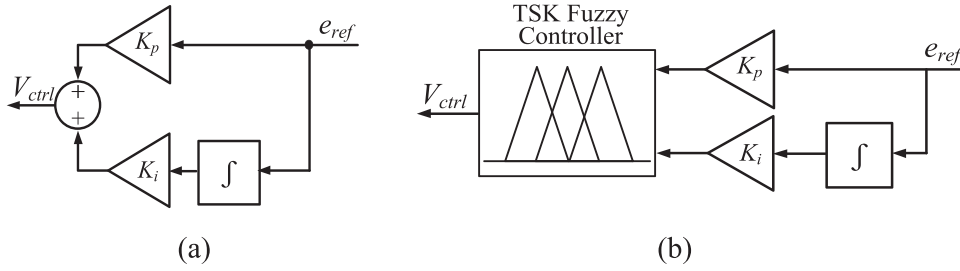


Fig. 3. Schematic model of a PI controller, (a) Classic, (b) Fuzzy.

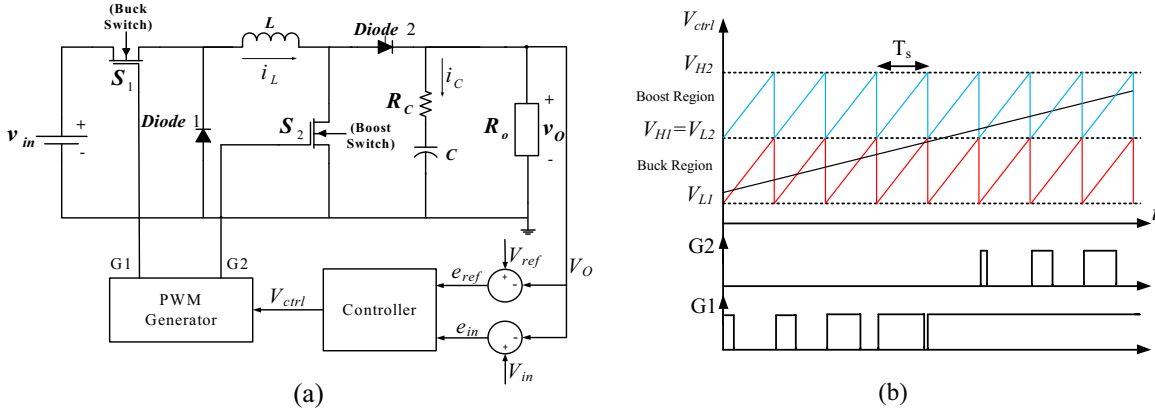


Fig. 4. The overall structure of a non-inverting buck-boost converter with control system. (a) The schematic model of a TLCS, (b) PWM generation strategy.

Table 3
Time-response characteristics and parameters for PI_{Buck} and PI_{Boost} controllers.

Parameters	Classic PI		Fuzzy PI	
	Buck	Boost	Buck	Boost
K_p	0.0163	0.0058	1.5000	1.0000
K_i	29.8603	15.0501	1.6000	15.0000
t_r (sec)	0.0092	0.0107	0.0051	0.0068
t_s (sec)	0.0169	0.0599	0.0099	0.0386
O.S. (%)	0	0	0	0
U.S. (%)	0	0	0	0

if the output voltage is greater than the input voltage, the switch must call a classic or a fuzzy PI_{Boost} . Furthermore, the fuzzy switch must be kept only one controller in the closed-loop control system at the time. The fuzzy switch has two inputs and two outputs. These two inputs are the difference between the input and output voltage (e_{in}) and the difference between the reference and output voltage (e_{ref}). Two outputs (Y_1, Y_2) determine which controller (PI_{Buck} or PI_{Boost}) must be inserted in the control loop. V_{ctrl} is the sum of PI_{Buck} and PI_{Boost} controllers outcomes.

Gaussian membership functions are used to describe the fuzzy switch inputs. Fig. 6 represents the first (e_{ref}) and the second (e_{in}) membership functions of the fuzzy switch inputs.

The following fuzzy rules are used to generate a TSK fuzzy switch. Fig. 7 demonstrates the decision surface of fuzzy rules.

Rule 1 IF e_{in} is Positive and e_{ref} is Negative THEN $y_1^1 = x_1 + x_2$ and $y_2^1 = x_1 + x_2$

Rule 2 IF e_{in} is Negative and e_{ref} is Positive THEN $y_1^2 = x_1 + x_2$ and $y_2^2 = x_1 + x_2$

Rule 3 IF e_{in} is Positive and e_{ref} is Positive THEN $y_1^3 = x_1 + x_2$ and $y_2^3 = x_1 + x_2$

Rule 4 IF e_{in} is Negative and e_{ref} is Negative THEN $y_1^4 = x_1 + x_2$ and $y_2^4 = x_1 + x_2$

In these rules x_1 and x_2 are the first and the second inputs, respectively. Fig. 8 depicts the realization of the proposed control scheme with the fuzzy switch.

The act of placing either a PI_{Buck} or a PI_{Boost} in the control loop is in fact the same as a decision making. Thus, the decision task is assigned to a TSK fuzzy system. Generally, TSK fuzzy systems are able to make either a soft or a hard decision. In this study the hard decision making which is the worst-condition in changing the controllers is considered. In hard decision making the two different dynamics controllers are instantly changed with each other, thus, a severe oscillation is exposed at the output voltage. Obviously, the oscillation may decrease if the TSK switch makes a soft decision in changing controllers. But even in a soft decision making, the oscillations are not completely cancelled out.

4.4. Modified two-level control scheme

A remarkable difference is observed in both classic and fuzzy PI_{Buck} and PI_{Boost} controllers output while there are replaced with each other at changing points. More precisely, in either of the classic or the fuzzy TLCS methods while one of the two PI_{Buck} and PI_{Boost} controllers is activated, the integral term of the either controller will be an arbitrary large value even if the controller zeros the error. Simply, transferring between two controllers, as shown in Fig. 5, leads a bump in the control signal resulting undesirable oscillations in the converter output voltage.

For instance, in Fig. 9(a), the output signals of classic PI_{Buck} and PI_{Boost} controllers through the different steps of reference voltage are presented. As long as the designed buck controller has larger gains (K_p and K_i), the control output is much larger than the boost controller. Similarly, there is a difference between the output of the fuzzy PI_{Buck} and PI_{Boost} controllers. Thus, replacing the controllers with each other at the changing points may cause fluctuations in the output voltage.

To solve the problem, the difference between outputs of two controllers must be minimized at the changing points to eliminate

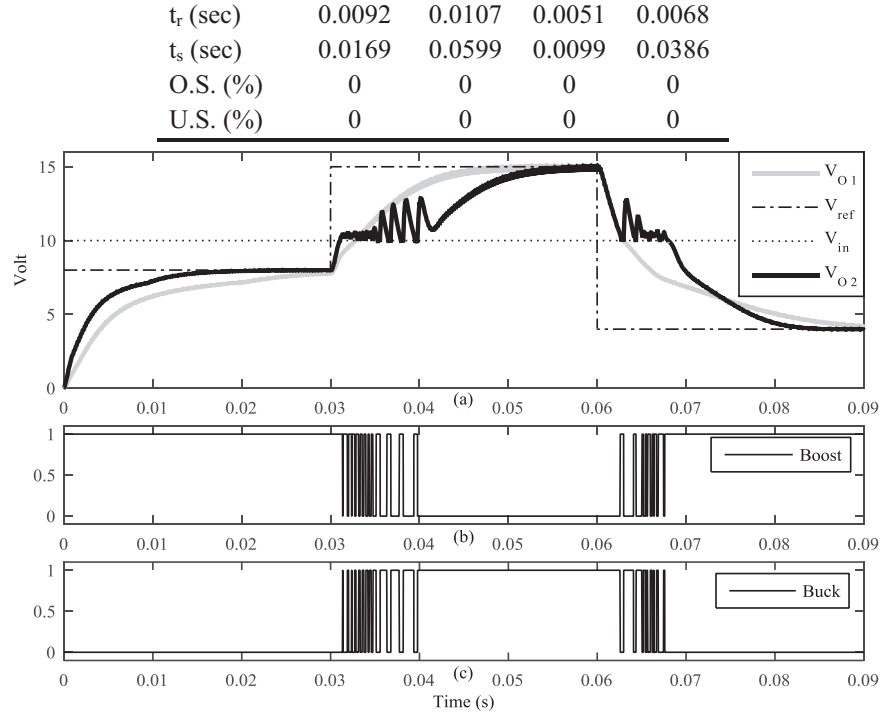


Fig. 5. Comparing the one-level PI controller output (V_{O1}) and the classic TLCS controller output (V_{O2}) with the fluctuation problem in the output voltage, (a) Output voltages, (b) Converter state in buck mode, (c) Converter state in boost mode.

the resultant oscillations in the output voltage. Researchers in [49–53], have addressed some solutions in industrial systems to cope with similar oscillation problem which is happened during the switching from automatic to manual mode and vice versa. Similarly, a solution for this problem have been introduced in the previous authors work to extend the output voltage range of a boost converter [30].

In this study, the solution is extended to resolve the switching oscillation problem at changing points in a classic and a fuzzy TLCS while driving a non-inverting buck-boost converter. Modifying the integration parts of classic and fuzzy PI controllers takes into account as the solution to this problem. First it should be considered that the main role of each PI controller is to minimize the existing error (e_{ref}) between V_{out} and V_{ref} . For this aim, an e_{bump} error signal which is obtained from the difference between the each PI con-

trollers outputs, i.e., PI_{Buck} , PI_{Boost} , and V_{ctrl} , is defined. Minimizing e_{bump} is added to the PI controllers tasks. As a consequence, the entire roles of PI controllers are to minimize both e_{bump} and e_{ref} at the same time. Note that whenever the difference between V_{ctrl} and the PI_{Buck} controller output is considered, e_{bump} is indexed by 1, i.e., e_{bump1} . And, similarly e_{bump} is indexed by 2 i.e., e_{bump2} , to show the existing difference between V_{ctrl} and the PI_{Boost} controller output. Then, both e_{bump1} and e_{bump2} are multiplied by their corresponding K_{b1} and K_{b2} gains with the same index numbering notification as the e_{bump} . These gains are defined the sharing effect of e_{bump} in the error signal. Then, the results of these multiplications are added to their corresponding integral term of PI_{buck} and PI_{Boost} controllers. There is no requirement to reset the integral part of each PI controller. Fig. 10 shows the schematic model of the modified classic and fuzzy PI controllers and Fig. 11 illustrates how

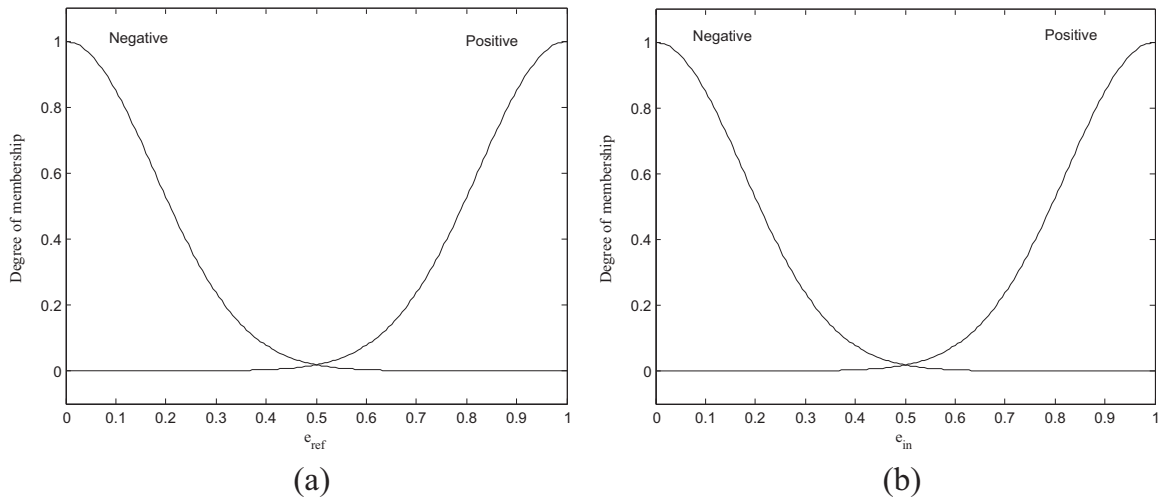


Fig. 6. Fuzzy switch input membership functions of, (a) First input (e_{ref}), (b) Second input (e_{in}).

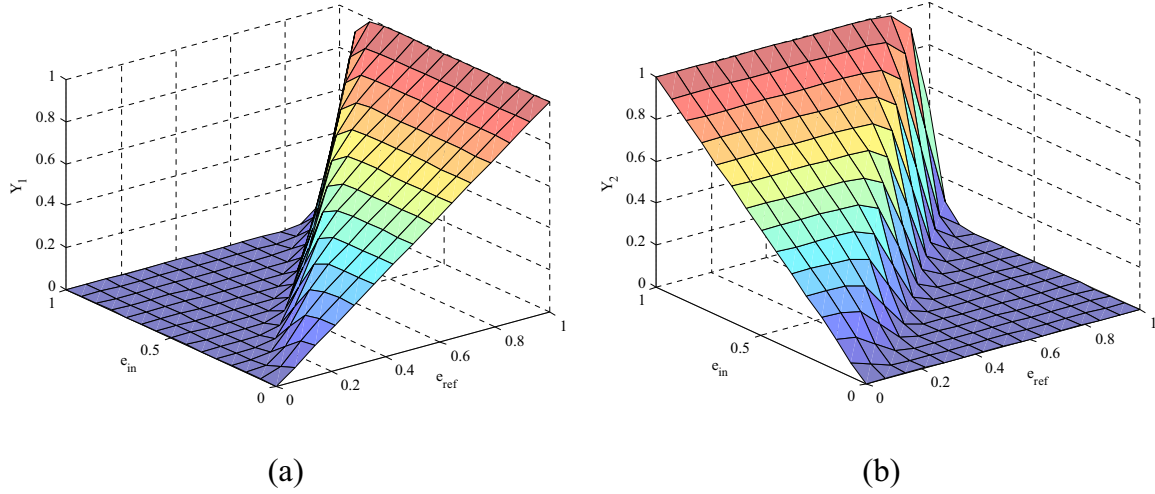


Fig. 7. Decision surface of a TSK fuzzy switch, (a) First output, (b) Second output.

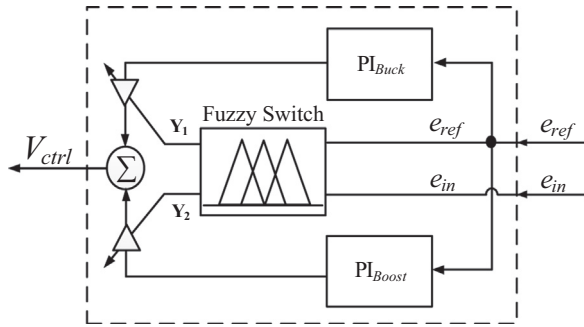


Fig. 8. The schematic model of a TLCS with a fuzzy switch.

a error signal e_{bump} is obtained and applied to the modified classic and fuzzy TLCS.

As a result of modifying the integral part of both classic and fuzzy PI_{Buck} and PI_{Boost} controllers, the outputs of controllers are closed together at the changing points. Unlike the previous situation (Fig. 5, no oscillation is generated while the classic PI_{Buck} and PI_{Boost} controllers are replaced with each other at changing points. Fig. 9(b) represents the modified classic PI_{Buck} and PI_{Boost}

controllers output.

5. Simulation and experimental results

In order to demonstrate the performance of both classic and fuzzy TLCSs, some simulations and experiments are conducted to control a non-inverting buck-boost converter. First, in simulation study, the superior efficiency of the modified classic TLCS over one-level controller is confirmed. Then the performance of modified classic and fuzzy TLCSs are compared. Because of the excellent outcome of the modified fuzzy TLCS, it is applied on the non-inverting buck-boost converter setup. The experimental results are provided to support the efficiency of the fuzzy TLCS.

The simulation result of the modified classic TLCS with the fuzzy switch during controlling the converter is drawn in Fig. 12. In this figure, V_{in} and V_{ref} are the input and the reference voltages, respectively. V_{O3} is the output voltage of the modified classic TLCS and V_{O1} is the output voltage of the one-level PI controller.

It should be noted that in the both classic and fuzzy TLCSs, the utilized PI_{Boost} controllers which are tuned for the optimal performance of the converter in the boost mode and have the same time-response parameters as the classic and the fuzzy one-level PI controllers. New classic and fuzzy PI_{Buck} controllers are adequately

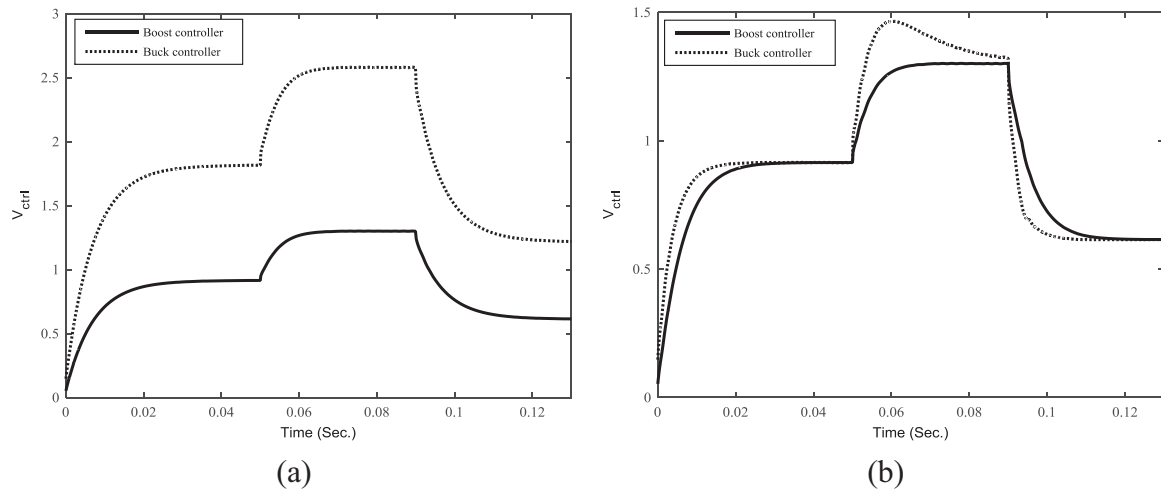


Fig. 9. The classic PI_{Buck} and PI_{Boost} controllers outputs in control loop of the non-inverting buck-boost converter, (a) Unmodified PI structure, (b) Modified PI structure.

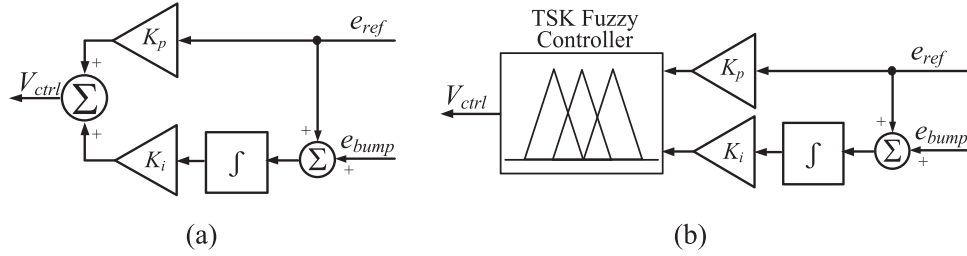


Fig. 10. Schematic model of the modified PI controller, (a) Classic, (b) Fuzzy.

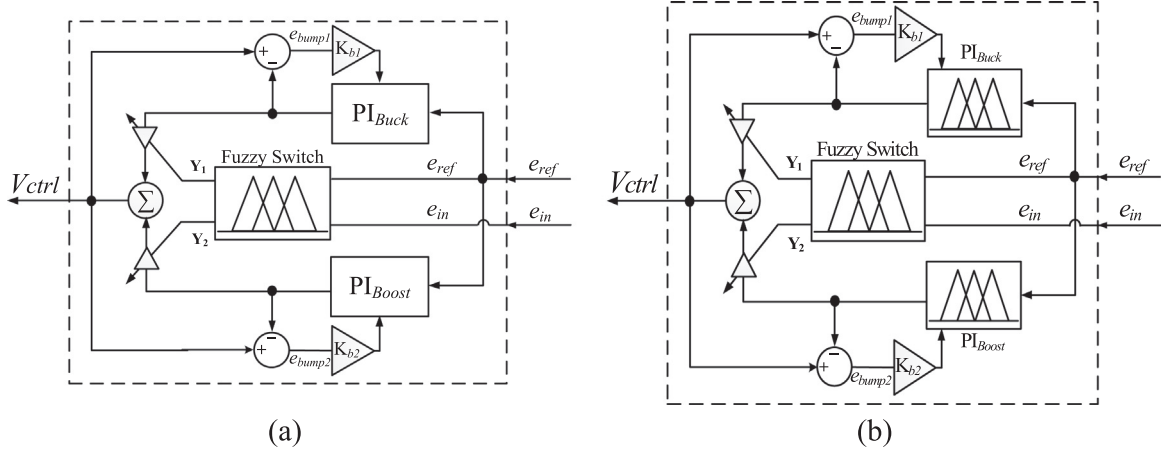


Fig. 11. Schematic model of the modified TLCS, (a) Classic, (b) Fuzzy.

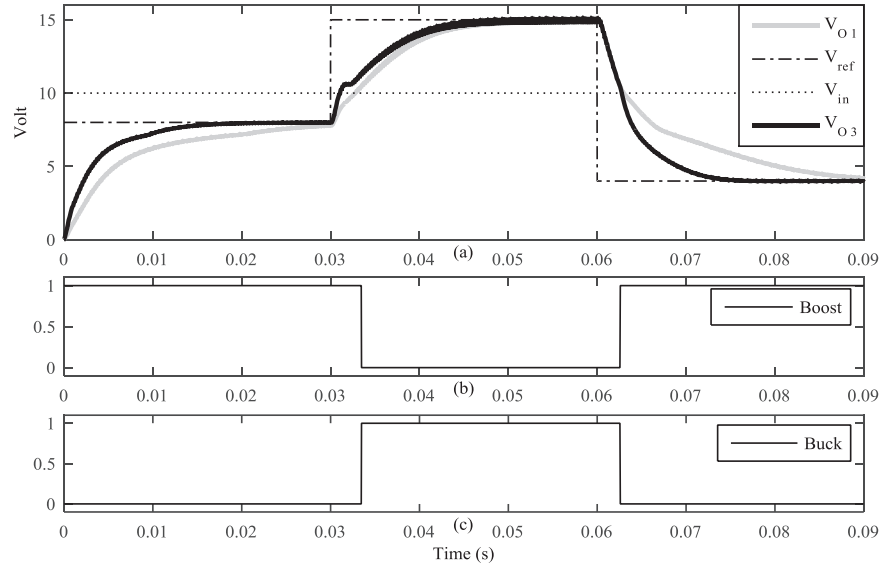


Fig. 12. Comparing the modified classic TLCS output (V_{O3}) and the one-level classic PI controller output (V_{O1}), (a) Output voltages, (b) Converter state in buck mode, (c) Converter state in boost mode.

designed for the buck operating mode of the converter in classic and fuzzy TLCS.

It is transparent that the TLCS in buck mode displays better performance in comparison with the one-level controller, because of using the properly designed PI_{Buck} controller in the buck mode of the converter. However, the behavior of the converter in boost mode in the two-level control design and the one-level controller is the same because of using an identical PI_{Boost} controller.

The two fuzzy PI controllers used in fuzzy TLCS are designed based on the method proposed in Section III. Each fuzzy PI controller is generated from the input–output data pairs of its corresponding classic PI controllers. Fig. 13 shows the membership

functions of a fuzzy PI controller.

The surface between inputs and outputs of fuzzy PI_{Buck} and PI_{Boost} are presented in Fig. 14.

The fuzzy control rules in form of Eq. (6) for the PI_{Buck} are as follows:

Rule 1) IF e is in1cluster1 and $\dot{e}$ is in2cluster1 THEN $y_1 = 0.01637 e + 30.86 \dot{e} + 4.125e-6$

Rule 2) IF e is in1cluster2 and $\dot{e}$ is in2cluster2 THEN $y_2 = 0.01637 e + 30.86 \dot{e} + 4.57e-6$

Rule 3) IF e is in1cluster3 and $\dot{e}$ is in2cluster3 THEN $y_3 = 0.01637 e + 30.86 \dot{e} + 4.10e-6$

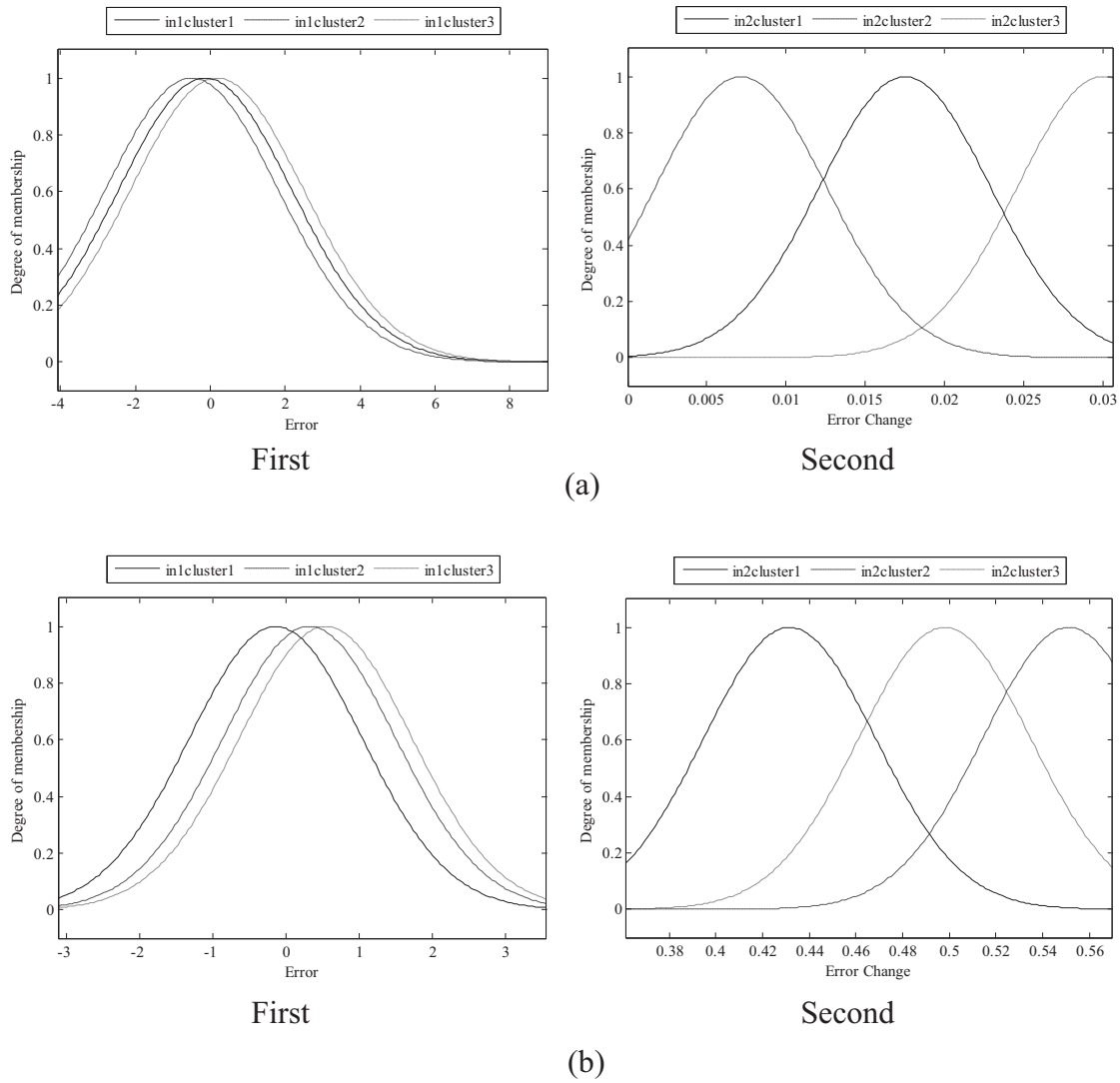


Fig. 13. The first (e) and the second ($\dot{e}$) Gaussian membership functions of fuzzy controllers, (a) PI_{Buck} , (b) PI_{Boost} .

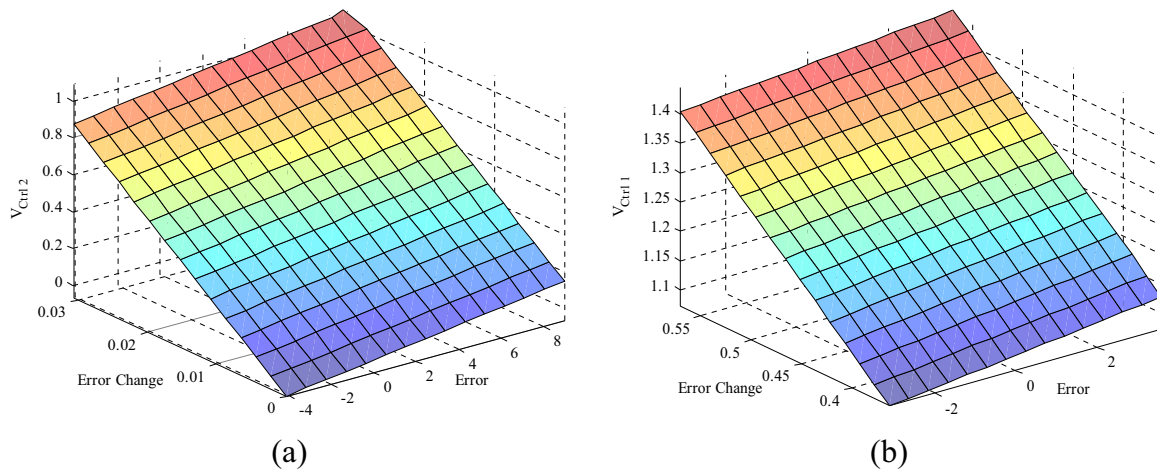


Fig. 14. The input–output surface of fuzzy PI controllers, (a) PI_{Buck} , (b) PI_{Boost} .

The fuzzy control rules in form of Eq. (6) for the PI_{Boost} are as follows:

Rule 1) IF e is in1cluster1 and $\dot{e}$ is in2cluster1 THEN $y_1 =$

$$0.008329 e + 1.597 \dot{e} + 0.5208$$

Rule 2) IF e is in1cluster2 and $\dot{e}$ is in2cluster2 THEN $y_2 =$
 $0.006017 e + 1.544 \dot{e} + 0.5438$

Rule 3) IF e is in1cluster3 and $\dot{e}$ is in2cluster3 THEN $y_3 =$

Table 4

Simulation parameters of a non-inverting buck-boost converter.

Description	Parameter	Value
Input voltage	V_{in}	10 V
Output voltage	V_o	8 V, 15 V, 4 V
Switching frequency	f_s	50 kHz
Inductance	L	50 μ H
Inductance ESR ^a	R_L	120 m Ω
Load resistance	R_o	2 Ω
Capacitance	C	1.8 mF
Capacitance ESR ^a	R_C	5 m Ω
Switches	S_1, S_2	IRF1407
Switch resistance	$R_{S1}=R_{S2}$	7.8 m Ω
Diode	Diode1,2	MBR20100CT
Diode resistance	$R_{D1}=R_{D2}$	40 m Ω
Sense input relation voltage		0.010822222222222*adc2 -2.313244444444439
Sense output relation voltage		0.005335195530726*adc0 +0.186340782122902

^a ESR: Equivalent Series Resistance.

$$0.007915 \text{ e} + 1.513 \text{ e} + 0.5545$$

Table 3 presents the time response characteristics parameters for both classic and fuzzy PI_{Buck} and PI_{Boost} controllers. The converter parameters used for both simulations and experiment tests are listed in Table 4.

The performance comparison between the modified classic and fuzzy TLCS is shown in Fig. 15.

The values of K_{b1} and K_{b2} gains for classic and fuzzy PI_{Buck} and PI_{Boost} controllers are shown in Table 5.

As illustrated in Fig. 15, the fuzzy TLCS provides better performance in comparison with the classic TLCS and the one-level controller. However, at the changing points of PI_{Buck} to PI_{Boost} (first step to second step of V_{ref}) and PI_{Boost} to PI_{Buck} (second step to third step of V_{ref}) replacement, no oscillations are founded in both classic and fuzzy TLCS. To observe the practical performance of fuzzy TLCSs, it is applied to an experimental setup. Fig. 16 shows the experimental setup comprising a 120 W non-inverting buck-boost converter, a DSP controller, and associated loads.

Two parallel diodes with 0.08 Ω resistance are used instead of a single diode in the test prototype; thus, their equivalent

Table 5The values of K_{b1} and K_{b2} gains.

Method	Mode	
	Buck (K_{b1})	Boost (K_{b2})
Classic	100	120
Fuzzy	25	1200

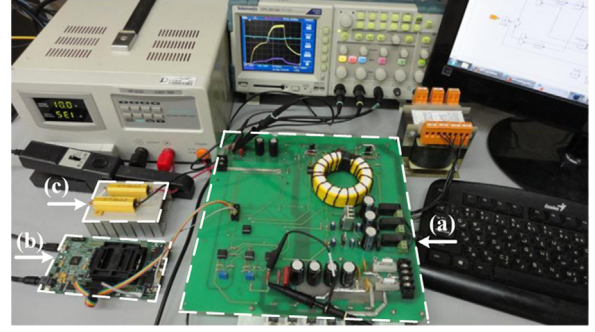


Fig. 16. Experimental setup, (a) non-inverting buck-boost converter, (b) DSP (eZdspTMS320F28335) controller, (c) Load.

resistances are considered in simulations. In addition, R_L is calculated by considering both the PCB tracks resistance and skin effect on resistance at 50 kHz switching frequency.

Fig. 17 provides the experimental waveforms when the reference voltage jumps from 8 V to 15 V and then falls to 4 V analogous to the simulations results in Fig. 15. The simulation and experimental results are in a good agreement. While Fig. 5 shows the fluctuation problem in unmodified TLCS output voltage at the changing points, the fuzzy TLCS as shown in Fig. 15(a) and Fig. 17 presents no oscillation during the changing modes.

While the transient time at the start-up is 30 ms for the one-level controller in Fig. 17(a), the fuzzy TLCS successfully reduces the transient time to about 11 ms in Fig. 17(b). Then, following to the reference voltage jumping from 8 V to 15 V, no noticeable difference in transient responses are identified. Because an identical PI_{boost} controller is being used in boost mode of the converter for both one level and fuzzy TLCS. At the final step, since reference voltage falls from 15 V to 4 V, the converter goes back to

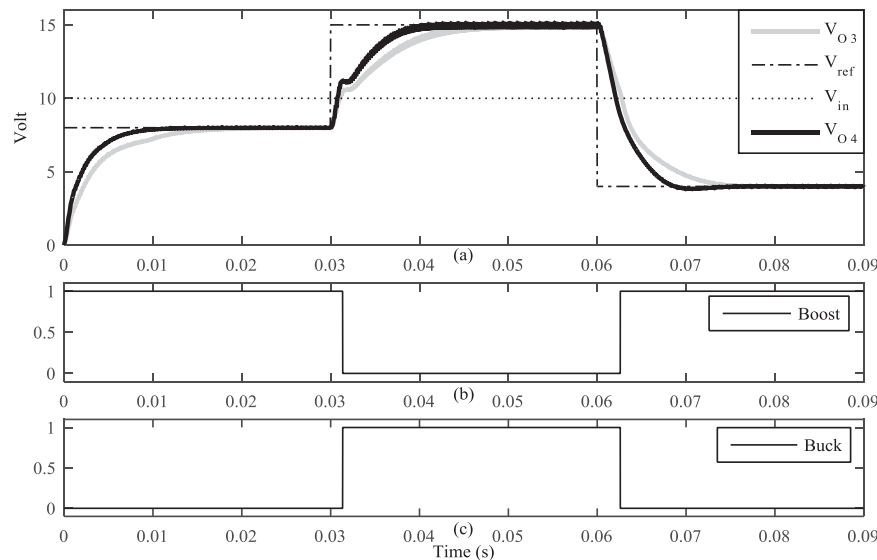


Fig. 15. Comparing the modified classic TLCS output (V_{O3}) and the modified fuzzy TLCS output (V_{O4}), a) Output voltages, b) Converter state of buck mode, c) Converter state of boost mode.

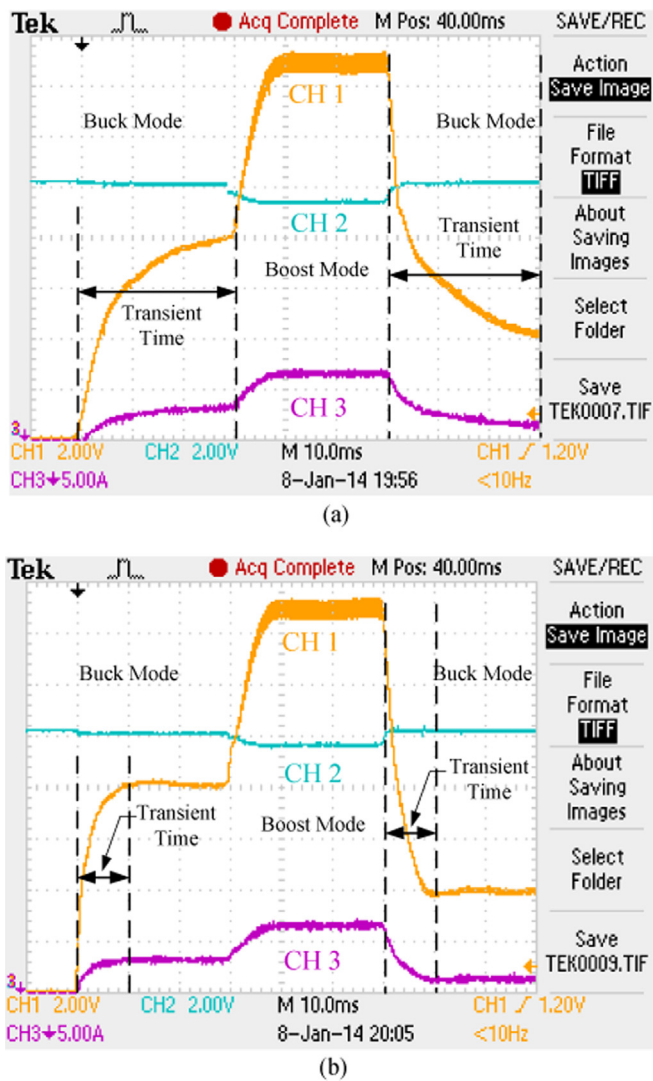


Fig. 17. Experimental waveforms of applying the modified fuzzy TLCS on the converter, (a) One-level PI controller, (b) Fuzzy TLCS. CH1: Output voltage, CH2: Input voltage, CH3: Output current.

the buck operating mode. In this case, while the comparison of buck modes transient time in Figs. 17(a) and (b) shows a significant difference, the modified fuzzy TLCS settling time is only for about one-third of the one-level PI controller.

Unfortunately, a PI controller itself is not a robust controller to deal with the possibility of existing uncertainties and nonlinearities in the system. The proposed method is a model-free based controller and it is not fair to compare our results with a model-based controller's outcome. However, in references [28,29], uncertainties and nonlinearities information of a system are considered in fuzzy membership functions to deal with parameter uncertainties of the system.

6. Conclusion

In this paper, a new control scheme was proposed to control the output voltage of a non-inverting buck-boost converter under variable reference voltages. In this control scheme, a fuzzy switch is responsible for placing either the modified fuzzy PI_{Buck} or PI_{Boost} controllers in control loop at certain moments proportional to the converter operating modes. The modification was applied to the structure of each fuzzy PI controllers to avoid occurring severe

transition oscillations during the control system switches from a fast-dynamic controller (fuzzy PI_{Buck}) to a slow-dynamic controller (fuzzy PI_{Boost}) and vice versa.

Both simulation and experimental results have confirmed that the proposed TLCS is effective in controlling the converter. Furthermore, according to the simulation and experimental results, the proposed TLCS has the capability of being used in controlling other converters with different dynamic operation modes. Also, the developed control scheme can be advanced by adding a multi-level controller for boost mode section in order to expand the operating range of the converter output. In addition, in our future work, we plan to expand our results to consider the possibility of nonlinearities and uncertainties in the system.

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